

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv(r"C:\Users\STUDENTS\Desktop\retail sale.csv")
print(df)
```

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	
0	1	11/24/2023	CUST001	Male	34	Beauty	
1	2	2/27/2023	CUST002	Female	26	Clothing	
2	3	1/13/2023	CUST003	Male	50	Electronics	
3	4	5/21/2023	CUST004	Male	37	Clothing	
4	5	5/6/2023	CUST005	Male	30	Beauty	
..	...	...	...	...	...	...	
995	996	5/16/2023	CUST996	Male	62	Clothing	
996	997	11/17/2023	CUST997	Male	52	Beauty	
997	998	10/29/2023	CUST998	Female	23	Beauty	
998	999	12/5/2023	CUST999	Female	36	Electronics	
999	1000	4/12/2023	CUST1000	Male	47	Electronics	

	Quantity	Price per Unit	Total Amount
0	3	50	150
1	2	500	1000
2	1	30	30
3	1	500	500
4	2	50	100
..	...	...	...
995	1	50	50
996	3	30	90
997	4	25	100
998	3	50	150
999	4	30	120

[1000 rows x 9 columns]

```
[1000 rows x 9 columns]
```

```
[3]: df.head()
```

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	11/24/2023	CUST001	Male	34	Beauty	3	50	150
1	2	2/27/2023	CUST002	Female	26	Clothing	2	500	1000
2	3	1/13/2023	CUST003	Male	50	Electronics	1	30	30
3	4	5/21/2023	CUST004	Male	37	Clothing	1	500	500
4	5	5/6/2023	CUST005	Male	30	Beauty	2	50	100

```
[4]: df.tail()
```

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
995	996	5/16/2023	CUST996	Male	62	Clothing	1	50	50
996	997	11/17/2023	CUST997	Male	52	Beauty	3	30	90
997	998	10/29/2023	CUST998	Female	23	Beauty	4	25	100
998	999	12/5/2023	CUST999	Female	36	Electronics	3	50	150
999	1000	4/12/2023	CUST1000	Male	47	Electronics	4	30	120

```
[5]: df.shape
```

```
[5]: (1000, 9)
```

```
[6]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype  
---  --
0   Transaction ID         1000 non-null  int64  
1   Date                  1000 non-null  object  
2   Customer ID           1000 non-null  object  
3   Gender                1000 non-null  object  
4   Age                   1000 non-null  int64  
5   Product Category      1000 non-null  object  
6   Quantity              1000 non-null  int64  
7   Price per Unit        1000 non-null  int64  
8   Total Amount          1000 non-null  int64  
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
```

```
[7]: df.describe()
```

	Transaction ID	Age	Quantity	Price per Unit	Total Amount
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	500.500000	41.39200	2.514000	179.890000	456.000000
std	288.819436	13.68143	1.132734	189.681356	559.997632
min	1.000000	18.00000	1.000000	25.000000	25.000000
25%	250.750000	29.00000	1.000000	30.000000	60.000000
50%	500.500000	42.00000	3.000000	50.000000	135.000000
75%	750.250000	53.00000	4.000000	300.000000	900.000000
max	1000.000000	64.00000	4.000000	500.000000	2000.000000

```
[8]: df.isnull().sum()
```

```
[8]: Transaction ID      0
Date                  0
Customer ID          0
Gender               0
Age                  0
Product Category     0
Quantity             0
Price per Unit       0
Total Amount         0
dtype: int64
```

```
[9]: df.duplicated().sum()
```

```
[9]: np.int64(0)
```

```
[15]: df.columns.tolist()
```

```
[15]: ['Transaction ID',
      'Date',
      'Customer ID',
      'Gender',
      'Age',
      'Product Category',
      'Quantity',
      'Price per Unit',
      'Total Amount']
```

```
[19]: print("Total Sales =", df["Total Amount"].sum())
Total Sales = 456000
```

```
[20]: print("Average Sales =", df["Total Amount"].mean())
Average Sales = 456.0
```

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Average Sales = 456.0

[21]: `print("Maximum Sales =", df["Total Amount"].max())`

Maximum Sales = 2000

[22]: `print("Minimum Sales =", df["Total Amount"].min())`

Minimum Sales = 25

[23]: `print("Total Customers =", df["Customer ID"].nunique())`

Total Customers = 1000

[24]: `print("Product Categories =", df["Product Category"].nunique())`

Product Categories = 3

[25]: `print("Total Quantity Sold =", df["Quantity"].sum())`

Total Quantity Sold = 2514

[26]: `df.groupby("Product Category")["Total Amount"].sum()`

[26]: Product Category
 Beauty 143515
 Clothing 155580
 Electronics 156905
 Name: Total Amount, dtype: int64

[27]: `df.groupby("Gender")["Total Amount"].sum()`

[27]: Gender
 Female 232840
 Male 223160
 Name: Total Amount, dtype: int64

```
[28]: df.groupby("Customer ID")["Total Amount"].sum()
```

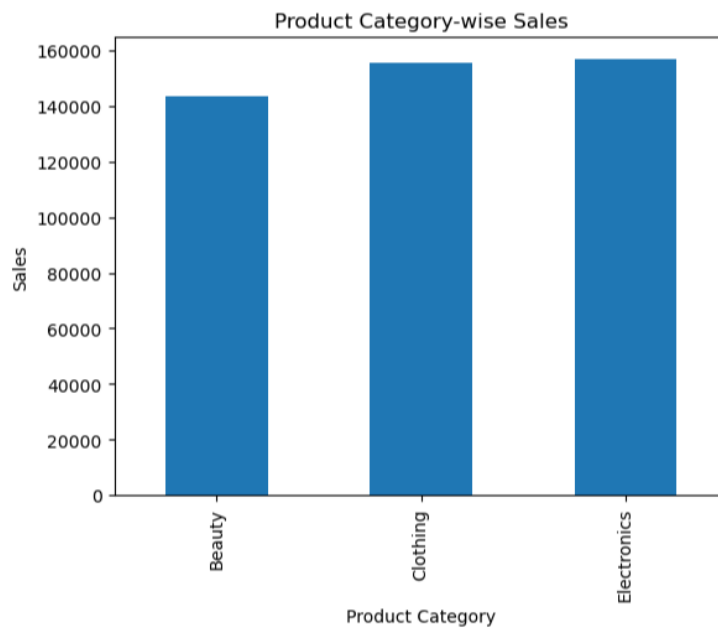
```
[28]: Customer ID
CUST001    150
CUST002   1000
CUST003     30
CUST004    500
CUST005    100
...
CUST995     30
CUST996     50
CUST997     90
CUST998    100
CUST999    150
Name: Total Amount, Length: 1000, dtype: int64
```

```
[29]: df.groupby("Product Category")["Total Amount"].sum().plot(kind="bar")

plt.title("Product Category-wise Sales")
plt.xlabel("Product Category")
plt.ylabel("Sales")
plt.show()
```

```
[29]: df.groupby("Product Category")["Total Amount"].sum().plot(kind="bar")

plt.title("Product Category-wise Sales")
plt.xlabel("Product Category")
plt.ylabel("Sales")
plt.show()
```



Product Category

```
[31]: df.groupby("Month")["Total Amount"].sum().plot(kind="line", marker="o")

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```



```
[32]: df.groupby("Product Category")["Total Amount"].sum().plot(kind="pie", autopct="%1.1f%%")

plt.title("Category-wise Sales")
plt.ylabel("")
plt.show()
```

